SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 5.01

C SAAT Mathematical Logic, Matrices & Complex Numbers

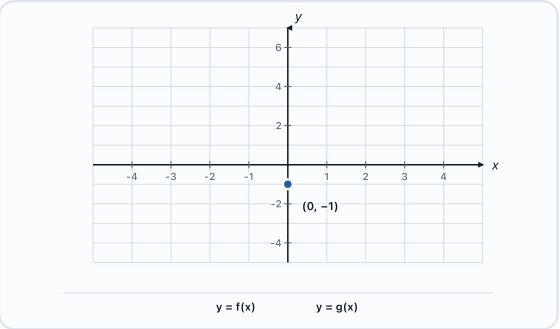
PRACTICE WORKSHEET

25 Structured Exam-Style Questions | Name: _____ | Date: _____ | Score: _____ / 25

SAAT Mathematical Logic, Matrices & Complex Numbers Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 14x + 64$ in the domain of SAAT Logic, Polynomials, Matrices & Complex Numbers. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(9)$.



A 34

B 41

C 19

D 32

2 MCQ

A mathematical model in SAAT Logic, Polynomials, Matrices & Complex Numbers is defined by the cubic function $y_2(x) = 3x^3 - 7x^2 + 15x - 12$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 96

B 54

C 39

D 28

3 MCQ

In SAAT Logic, Polynomials, Matrices & Complex Numbers, two displacement vectors are given by $\mathbf{u}_3 = \langle 7, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A 7

B 113

C 120

D 232

4 MCQ

The linear growth profile shown in the diagram for SAAT Logic, Polynomials, Matrices & Complex Numbers follows an arithmetic progression with first term $a_1 = 8$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 92

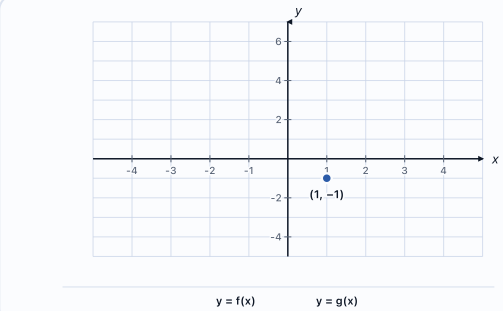
B 840

C 1380

D 750

5 MCQ

In SAAT Logic, Polynomials, Matrices & Complex Numbers, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 28x)\,\mathrm{d}x$.



A 104

B 128

C 132

D 53

6 MCQ

In SAAT Logic, Polynomials, Matrices & Complex Numbers, consider the quadratic function $f_6(x) = x^2 - 16x + 151$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(15)$.

A 231

B 223

C 136

D 216

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 8x^2 + 33x - 19$ studied in SAAT Logic, Polynomials, Matrices & Complex Numbers. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 609

B 448

C 481

D 89

8 MCQ

In SAAT Logic, Polynomials, Matrices & Complex Numbers, two displacement vectors are given by $\mathbf{u}_{8} = \langle 8, -7 \rangle$ and $\mathbf{v}_{8} = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_{8} \cdot \mathbf{v}_{8} + \|\mathbf{u}_{8}\|^{2}$.

A 29

B 113

C 212

D 142

9 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 2 & 3 \\ 6 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 6

B 9

C -9

D 11

10 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 3 & 4 \\ 12 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 16

B -16

C 12

D 19

11 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 4 & 5 \\ 20 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A -25

B 20

C 25

D 29

12 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 5 & 6 \\ 30 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 30

B 36

C 41

D -36

13 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 6 & 7 \\ 12 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 20

B 14

C 42

D -14

14 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 7 & 1 \\ 21 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A -3

B 3

C 10

D 7

15 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 2 & 2 \\ 8 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 10

B 8

C -8

D 4

16 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 3 & 3 \\ 15 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 9

B -15

C 18

D 15

17 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 4 & 4 \\ 24 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 28

B 16

C -24

D 24

18 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 5 & 5 \\ 10 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 15

B -10

C 10

D 25

19 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 6 & 6 \\ 18 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 36

B 18

C 24

D -18

20 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 7 & 7 \\ 28 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 49
B 35
C -28
D 28

21 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 2 & 1 \\ 10 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 2
B -5
C 7
D 5

22 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 3 & 2 \\ 18 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 12
B -12
C 15
D 6

23 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 4 & 3 \\ 8 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 6
B -6
C 12
D 10

24 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 5 & 4 \\ 15 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 12
B 20
C 17
D -12

25 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 6 & 5 \\ 24 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A -20
B 20
C 30
D 26